Comprehension Exercise

Working With Electricity

Answers

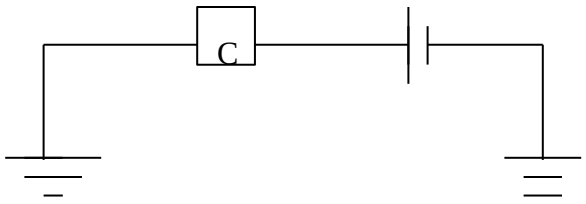
1 The screen becomes charged due to the bombardment of electrons on it over a period of time. The dust particles which may be negatively charged or neutral are attracted to the screen.

2 If there is a build up of charge on the tanker it will be conducted to the earth and not build up to the stage where it will cause a spark and hence an explosion. A build up of charge could be caused by the fuel running through the pipe.

3 Negative charges will be attracted to neutral surfaces as well as positive surfaces.

4 Charge will "jump" to either a neutral point or a point of opposite charge. The door handle does not have to be charged as long as the person is carrying a charge.

5



6 The water would be chemically separated into hydrogen and oxygen. The process is called electrolysis.

7 The plastic container (being and insulator) may build up an electric charge on its surface. If there is a discharge spark the fuel may explode. The charge can be built up by the petrol moving within the container or the outside of the container rubbing against something.

8 $Q = I \times t$

$Q = (20 \times 10^{-3}) \times 0.1$

$= 20 \times 10^{-4}$ coulombs